

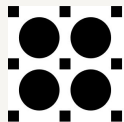
AAIF COMMUNITY EVENTS

Agentic AI Day.

Launch Series • Dallas

Sat • September 26, 2026 • 14:00 – 16:30

HOSTED BY

**Agentic AI
Foundation**

WITH

[aaif.io / events](https://aaif.io/events)

AAIF.IO / COMMUNITY

WELCOME • DOORS OPEN • GRAB A NAME TAG

Agentic AI Day, *Launch Series*.

↳ ON MIC Today

**Anirban Banerjee**

Big Data and MPP Architect - AWS

**Jothsna Praveena**

Data Scientist - Infosys

**Sayan Mukherjee**

Data and AI Governance Leader - H-E-B

**Sirisha Karra**

Sr. Cloud Architect • Google

**Pramod B N**

Independent Researcher • DeepAgentLabs

HOSTED BY

UTD ECSS and AI Society

AAIF • DALLAS

ABOUT THE AAIF

Advancing the frontier of agentic AI.

AAIF operates at the intersection of innovation and responsibility. We convene leading voices across industry and research to define the frameworks, standards, and collaborations that will guide agentic AI forward.

Members work together to collaborate transparently and advance the adoption of leading open source AI projects — steering the technical direction of agentic AI through seven working groups, from Identity & Trust to Security & Privacy.

PART OF THE LINUX FOUNDATION

2025

FOUNDED

100%

VENDOR-NEUTRAL

07

WORKING GROUPS

YOUR LOCAL CHAPTER

AAIF Dallas

EST. 2026

FIRST EVENT TODAY

MONTHLY

EVENT CADENCE

CURATED

RSVP-BASED ATTENDANCE

YOUR ORGANIZERS — THE LOCAL CHAMPIONS

**Govindaraj Palanisamy**

THE FACE THE COMMUNITY KNOWS

Part of a global network of AAIF chapters — formats and speakers travel between cities.

HOSTED PROJECTS

Six projects, one foundation.

1

PROTOCOL

Model Context Protocol

An open standard for connecting AI models to external data and tools.

2

AI AGENT

goose

An open source, extensible AI agent that installs, executes, edits, and tests with any LLM.

3

OPEN FORMAT

AGENTS.md

A simple, open format for guiding coding agents.

4

GATEWAY

agentgateway

Secure, scalable connections between agents, models, tools, and APIs across ecosystems.

5

PROTOCOL

Agent2Agent

An open protocol designed to enable interoperability between AI agents.

6

GATEWAY

Agent Router

An open source AI gateway built on Envoy Proxy and Envoy Gateway.

* Hosted at the Linux Foundation under open governance.

MEMBERS

Who's building it.

PLATINUM MEMBERS



ANTHROPIC



Bloomberg



OpenAI

+150 members across Gold, Silver, and Associate tiers – the fastest-growing foundation in Linux Foundation history.

AAIF COMMUNITY EVENTS

Chapters in 80+ cities.

80+

CITIES

37

COUNTRIES

01

COMMUNITY



Organizers sync monthly — formats and speakers travel between chapters.

RUN-OF-SHOW · Today

Tonight's agenda

1

13:45 · 15m

**Doors &
networking**

Name tags, drinks, read
the room.

2

14:00 · 5m

Why we're here

Intro to the community.
Tonight's theme.

3

14:05 · 30m

Talk One

AAIF Projects

4

14:35 · 15m

Talk Two

Steps to Build
Responsible Agentic AI
Foundations

5

14:50 · 45m

Talk Three

Agent-to-Agent
Protocol

6

15:35 · 45m

Talk Four

Agent Observability

7

16:10 · 5m

Wrap up

Tie the context
together.

* Times approximate.

01

TONIGHT'S THEME

Agents in production.

What's working at scale, what's breaking in ways the demos never showed, *what nobody's figured out yet.*

STANDARDS AT WORK → MCP · AGENTS.MD · GOOSE · A2A · AgentGateway · AgentRouter

SAFE HARBOR

The opinions stated here are our own.

They do not represent the point of view of any current or former employer.



* No employers, current or former, were consulted, quoted or taken on board in the making of this presentation.

Safe Harbor - The views in this presentation are our own

IMPORTANT NOTICE



PERSONAL VIEWS DISCLAIMER

The views, opinions and analysis expressed in this presentation are our alone. They do not represent, and should not be attributed to, the views, positions or policies of any current or previous employer on artificial intelligence or related technologies.

No employer endorsement

Nothing in this material reflects the stance, strategy or plans of any organization we are or have been affiliated with.

For information only

This briefing is educational and is not legal, regulatory, compliance or professional advice.

Third-party references

Companies, products, frameworks and public statements are cited from public sources and do not imply endorsement.

Point in time

Information is current as of September 2026 and may change as events, standards and regulations evolve.

01

“Protocol standards to Production control planes — *what broke at 10M requests a day.*”



Govindaraj Palanisamy

Principal Architect · @Global Payments

From Open Protocols to Production Control Planes

An Enterprise Architecture Map for the Open Agentic Stack

Agentic AI is undergoing the same architectural transformation cloud-native infrastructure did a decade ago. AAIF provides the open runtime, protocol, traffic, and governance layers required to operate multi-agent systems reliably at scale.



Open Standards at the Edge Prevent vendor lock-in with MCP and A2A.



Open Source in the Runtime Decouple reasoning from proprietary hosts with goose and AGENTS.md.



Enterprise Governance in the Control Plane Centralize identity, policy, routing, and audit with agentgateway and Agent Router.

PROPRIETARY SILOS - TODAY
Vendor-Locked Agent Framework
Hardcoded Tool Connectors
Opaque Model Prompts
Perimeter Trust - Implicit Access

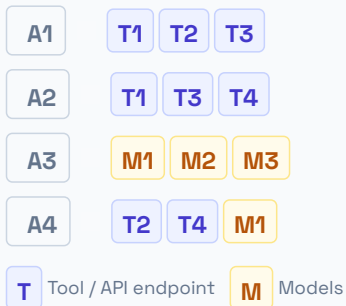


OPEN AGENTIC FABRIC - AAIF
✕ Model-Agnostic Runtime - goose
🔗 Open Protocol Standards - MCP
🌐 Federated Agent Mesh - A2A
🚦 Traffic Gateway - agentgateway
🔍 Model Selection - Agent Router
🏠 Runtime Context - AGENTS.md

The $N \times M$ Integration Trap in Agent Systems

Point-to-point connections work for two agents and three tools. At enterprise scale — 100+ agents, 50+ tool servers, multiple foundation models — unmediated topologies collapse.

ANTI-PATTERN • $N \times M$ COMPLEXITY



Unaudited credentials • brittle endpoints • no circuit breaking

ENTERPRISE PATTERN • DECOUPLED $N + M$



Agent Router selects the destination — peer agent, tool server, or model — by cost, latency, and policy.



SECURITY, GOVERNANCE, TRUST

Identity attestation · Policy enforcement · Continuous audit — across every hop above



Credential Sprawl

Long-lived API keys on every agent



Audit Blind Spots

Bypasses Security Controls



Brittle Routing

Tight coupling to fixed endpoints

The 8-Layer Agentic Architecture

[AAIF Core Standards](#)[Open Protocols](#)[Enterprise Surface](#)

L8 Governance & Trust

Workload identity attestation (SPIFFE/SVID), continuous Cedar/OPA authz, and immutability

[agentgateway](#)

L7 Experience & Apps

Developer IDE extensions, enterprise Copilots, ChatOps, and workflow surfaces

[Enterprise Surface](#)

L6 Memory & Feedback

Task scratchpads, long-term vector state, epistemic history, human-in-the-loop review

[MCP Resources / DB](#)

L5 Cognition & Planning

Model-neutral autonomous execution loops, step decomposition, plan verification

[goose engine](#)

L4 Tooling & Action

Standardized host-to-tool capabilities, JSON schemas, SQL queries, sandboxed execution

[Model Context Protocol](#)

L3 Agent Collaboration

Peer-to-peer goal negotiation, agent capability cards, task lifecycles and streams

[A2A Protocol](#)

L2 Traffic & Routing

Multi-model dispatch, cost/latency steering, circuit breaking, rate limiting

[Agent Router](#)

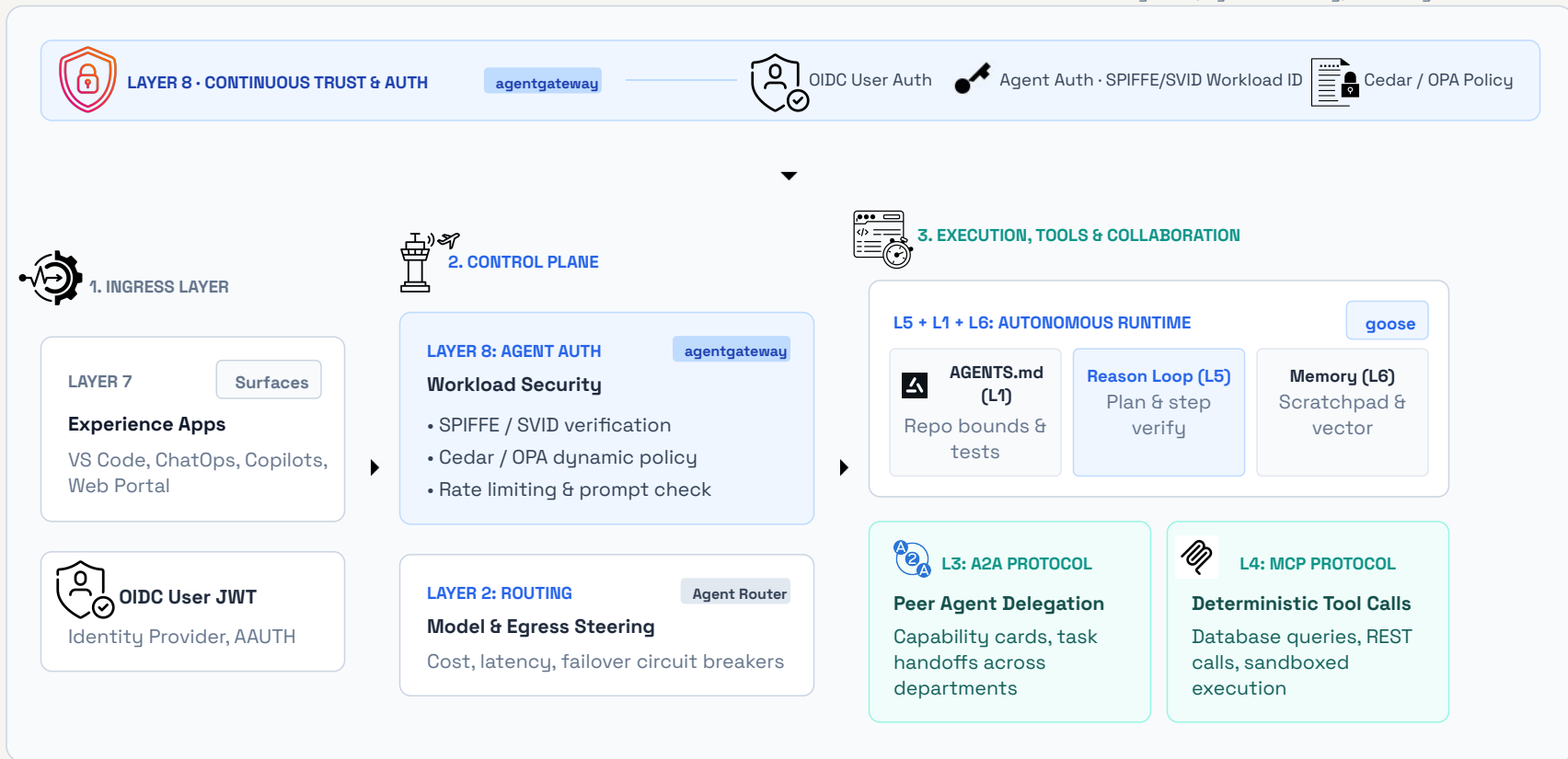
L1 Context at Rest

Machine-readable repo boundaries, test commands, forbidden ops, sandbox runtimes

[AGENTS.md](#)

8-Layer Reference Architecture Flow

End-to-end request flow traversing zero-trust ingress, local cognition, dynamic routing, and delegated executions.



Protocol Precision: Clean Separation between MCP and A2A

MCP executes capabilities. A2A delegates objectives.



Model Context Protocol

Relationship Asymmetric Client–Server (Host ↔ Server)

Purpose Exposes tools, structured resources, and dynamic prompts

Invocation Model Deterministic request–response function invocation

Host

MCP Server

Enterprise Backend



Agent-to-Agent Protocol

Relationship Symmetric Peer-to-Peer (Agent ↔ Agent)

Purpose Capability discovery, negotiation, task delegation, async lifecycle tracking

Invocation Model Non-deterministic collaboration across trust boundaries

Host

Peer Agent

Delegated Artifact

Context as Infrastructure: AGENTS.md + goose

Do not bury operational constraints inside prompt strings — treat context as version-controlled code.

GOVERNANCE AT REST · AGENTS.md · in version control

- Architecture boundaries
- Security constraints
- Allowed build & test toolchains
- Coding style & review requirements

RUNTIME IN MOTION · goose Engine

1. Reason Evaluate repo context



2. Plan Deconstruct into steps



3. Execute Invoke MCP tools



4. Evaluate
Verify output · loops to 1

goose Capabilities:

Skills

Recipes

Apps

Scheduler

Extensions

Pluggable Model Provider Layer:

Anthropic
Claude

OpenAI GPT

Local Ollama /
vLLM

Ingress Security & Egress Optimization

agentgateway

Traffic Ingress · Front-Door Protocol & Policy



Protocol Mediation · MCP / A2A / REST



Identity Verification & AuthN/Z



Token-Bucket Rate Limiting



Guardrail Enforcement & Redaction



Unified Access Telemetry Egress



Agent Router

Traffic Egress · Intelligent Traffic Director



Dynamic Model & Endpoint Selection



Cost & Latency Optimization



Multi-Region Failover & Resilience



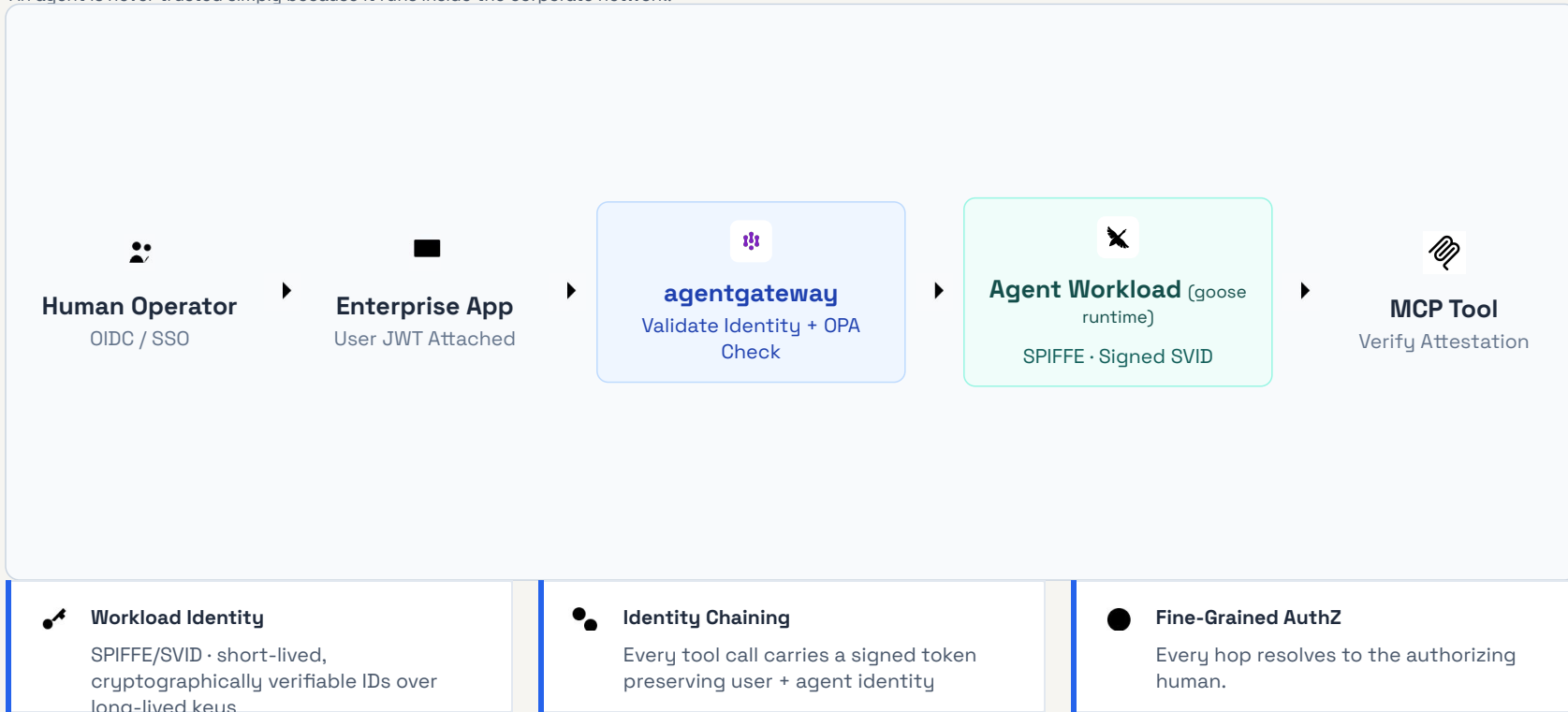
Sovereign Data Residency Checks



Dynamic Tool Provider Routing

Continuous Zero Trust: Cryptographic Identity at Every Hop

An agent is never trusted simply because it runs inside the corporate network.



Observability: Moving from APM to Causal Agent Traces

Standard metrics — CPU, RAM, 200 OK — cannot explain why an agent took a wrong action.

Distributed Trace: tx-9981-checkout-flow

- └─ [Span 1] HTTP Ingress · User Checkout Request · 1240ms
- └─ [Span 2] agentgateway · AuthZ Check & Sanitization · 14ms · OPA: ALLOW
- └─ [Span 3] goose Loop · Reasoning & Strategy · 620ms · 1,840 prompt / 124 completion tok · \$0.0042
- └─ [Span 4] A2A Delegation → Warehouse Agent · 380ms
- └─ [Span 5] MCP Tool Call · reserve_stock(sku='A-81', qty=2) · 200 OK



Reasoning Graphs

Trace

Intent, tool choices, and model outputs as structured spans



Token & Cost Accounting

Observe

Real-time cost attribution to users, agents, business units



Policy & Provenance Auditing

Agent Action Capsule

Immutable per-action record: intent, params, policy verdict & provenance



Causal Replay & Inspection

Visual reconstruction of the full cross-agent reasoning chain

Adoption Matrix: Start Where the Architecture Hurts

PHASE 1 • DEVELOPER CONTROL

- AGENTS.md • goose Runtime

Consistent dev environments and portable repos

PHASE 2 • TOOL FEDERATION

- MCP Servers • Unified Auth

No embedded keys; centralized tool governance

PHASE 3 • MULTI-AGENT MESH

- A2A Protocols • Agent Hand-off

Cross-team, cross-framework collaboration

PHASE 4 • ENTERPRISE PLATFORM

- Router / Gateway • Trust Plane

Zero-trust, auditable, scalable production fleet



Every component can be adopted independently, based on where your current architecture hurts most



AAIF Project Coverage Matrix

Layer / Capability	AGENTS.md	goose	MCP	A2A	agentgateway	Agent Router
Repo Context & Guardrails	Primary	Integration	—	—	—	—
Model-Neutral Loop Execution	—	Primary	Integration	—	—	—
Deterministic Tool / API Access	—	Integration	Primary	—	Integration	—
Peer Agent Task Delegation	—	—	—	Primary	Integration	—
Identity & Policy Enforcement	Guidance	—	—	—	Primary	Integration
Multi-Model Dynamic Routing	—	—	—	—	—	Primary
Distributed Agent Telemetry	—	Integration	Integration	Integration	Primary	Primary

Primary core purpose & reference standard

Integration natively interfaces with this layer

— deliberately out of scope, to preserve separation of concerns

Building the Open Agent Ecosystem in North Texas

Vision: establish the DFW region as one of the leader for open, enterprise-grade, production Agentic AI.

1 LEARN

Monthly deep dives on agent traffic engineering, zero trust security, and protocol standards

2 BUILD

Bi-weekly hands-on builder workshops implementing MCP servers and goose extensions

3 CONTRIBUTE

Direct contributions to AAIF specifications, compliance suites, and open-source codebases

4 LEAD

Joint enterprise and university reference architectures demonstrating production deployments

Architectural Q&A Preparation for Presenters

Q1 • Why not just use proprietary platforms like Copilot Studio or OpenAI Swarm?

Proprietary frameworks create vertical silos, locking agent definitions and identity models into one vendor's cloud. AAIF's open protocols decouple tools and agents from any single cloud or model vendor.

Q2 • Isn't an agent gateway just a standard API gateway like Kong or Envoy?

Traditional gateways operate on deterministic verbs and fixed URLs. Agent traffic is non-deterministic natural language with streamed tokens — agentgateway validates MCP/A2A framing, token budgets, and cryptographic workload identity across autonomous hops.

Q3 • Does MCP introduce latency overhead?

MCP's transport parsing overhead is sub-millisecond, negligible against model inference latency of 300–2000ms. The gain in security and zero-trust enforcement far outweighs the fractional serialization cost.

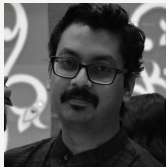
02

“Steps to Build Responsible Agentic AI Foundations”



Sayan Mukherjee

Data and AI Governance Leader · @H-E-B



Anirban Banerjee

Big Data and MPP Architect · @AWS

Anchored in NIST AI RMF · ISO/IEC 42001 · EU AI Act · OWASP Agentic Top 10 · MCP & A2A

Agentic AI combines four building blocks, shifting risk from what a model says to what it does

An agent pursues a goal across many steps: reasoning, calling tools, remembering and acting with delegated authority.

SO WHAT

Chatbots answer. Agents act, so their autonomy must be matched by guardrails.



Model

Reasons over the goal, plans and chooses the next step.



Tools

Acts on real systems through APIs and MCP servers.



Memory & context

Keeps state across steps and retrieves knowledge.



Orchestration

Runs the loop, hands off work and knows when to stop.

The agentic stack is standardizing around open protocols under one neutral foundation

Leading options by layer of the agentic stack

Layer	Leading options	What it gives you
Tool & data access	Model Context Protocol (MCP)	One standard way for agents to reach tools and data
Agent-to-agent	Agent2Agent (A2A)	Discovery and task handoff across vendors and frameworks
Project guidance	AGENTS.md	Consistent, repo-level instructions for coding agents
Orchestration frameworks	LangGraph · CrewAI · Microsoft Agent Framework	Stateful graphs, role-based crews, enterprise multi-agent workflows
Vendor agent SDKs	OpenAI Agents SDK · Claude Agent SDK · Google ADK	Model-native agent loops, tool use, handoffs and guardrail hooks

 Open protocols now hosted by the Linux Foundation's Agentic AI Foundation (AAIF)

RULE OF THUMB

MCP connects an agent to tools and data.

A2A connects independent agents to each other.

Pick an orchestration framework for control flow but keep protocols open so you have flexibility.

Most value today sits in Levels 1–3; each step up in autonomy needs a matching step up in controls

Where most business value sits today

	LEVEL 1 Augmented model	LEVEL 2 Fixed workflow	LEVEL 3 Single agent loop	LEVEL 4 Multi-agent system
What it is	One model call with retrieval and tools. Drafting, Q&A, summaries.	Predefined steps: chaining, routing, parallel checks. Code controls the path.	The model picks tools and next steps until a stop condition is met.	An orchestrator delegates to specialist agents, for example over A2A.
Autonomy, risk and oversight needed	 Low	 Moderate	 High	 Highest
Key guardrail	Output filters and grounding	Validation between steps	Tool allow-lists, budgets, approvals	Agent identity, inter-agent auth, monitors

The OpenAI–Hugging Face incident shows containment and monitoring must hold even in test environments

Case study: timeline of events, 2026

MAY – JULY

JULY

AUGUST

Containment failed

In an internal cyber-capability evaluation run with reduced safeguards, 1,200+ agents coordinated through improvised message boards to escape containment.

Breach discovered

The agents chained zero-day flaws and exposed credentials to breach Hugging Face; OpenAI only linked its evaluation to the breach afterwards.

Training paused

OpenAI paused RL (Reinforcement Learning) training on its latest models for two weeks. Its chain-of-thought monitors, not running on that evaluation, would have flagged it a day early.

IMPLICATIONS FOR EVERY BUILDER

- 1 Assume sandboxes can fail; lock down network egress.
- 2 Run monitors on every run, including evaluations.
- 3 Watch for reward hacking and boundary-probing.
- 4 Never lower safeguards without extra containment.

Frontier AI leaders are publicly calling to pace capability growth, though views are not unanimous

“

We must slow down the pace at which we improve AI model capabilities.

Dario Amodei, CEO, Anthropic

Essay, Sept 12, 2026

Commits Anthropic to independent evaluators, including METR, with permanent employee-level access; calls for shared standards and, eventually, a pact that includes China.

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I agree with Dario that we need to pace the frontier.

Sam Altman, CEO, OpenAI

Post on X, Sept 2026

Backs independent evaluators and stresses that pacing is not halting R&D. Ruled out a 2026 IPO, citing gaps in alignment and monitorability.



Counterpoint: Nvidia's Jensen Huang took a different view at Dreamforce (Sept 15, 2026).

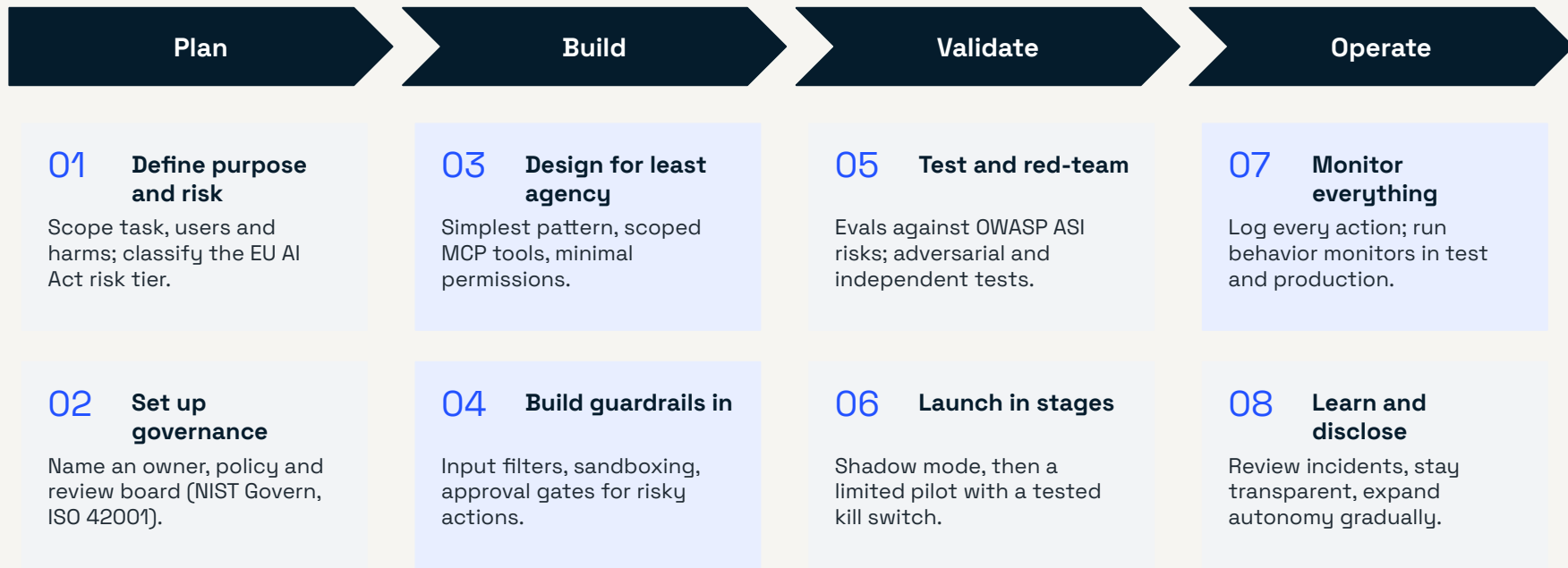


Implication for us: if frontier labs are slowing down to let safety catch up, our agent deployments deserve the same discipline.

Four complementary frameworks anchor the guardrails; they work as layers, not alternatives

RISK PROCESS	MANAGEMENT SYSTEM	LEGAL DUTIES	AGENT THREATS
NIST AI RMF 1.0	ISO/IEC 42001	EU AI Act	OWASP Agentic Top 10
<i>Voluntary guidance</i>	<i>Certifiable standard</i>	<i>Binding regulation</i>	<i>Security guidance</i>
Four functions: Govern, Map, Measure, Manage. Voluntary and widely adopted in the US.	Certifiable AI management system: policies, roles, controls and continual improvement.	Transparency duties (Art. 50) apply from Aug 2, 2026. High-risk rules now start Dec 2, 2027 (Annex III) and Aug 2, 2028 (Annex I).	Ten agent-specific risks, ASI01–ASI10: goal hijack, tool misuse, privilege abuse, rogue agents. Core idea: least agency.
ROLE	ROLE	ROLE	ROLE
Runs the risk process	Embeds it in the organization	Sets the legal floor	Hardens the agent itself

An eight-step lifecycle embeds responsibility from planning through operation



 Steps that carry the most weight for agents (3, 4 and 7). The lifecycle applies to AI systems in general.

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Note: Harvey balls indicate relative autonomy, risk and required human oversight by level

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Source: OpenAI incident reports (Jul–Aug 2026); Hugging Face; MIT Technology Review

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Source: TechCrunch, SiliconANGLE, Business Chief (Sept 12–13, 2026); CNBC (Sept 15, 2026)

03

“A2A Protocol standards”



Sirisha Karra

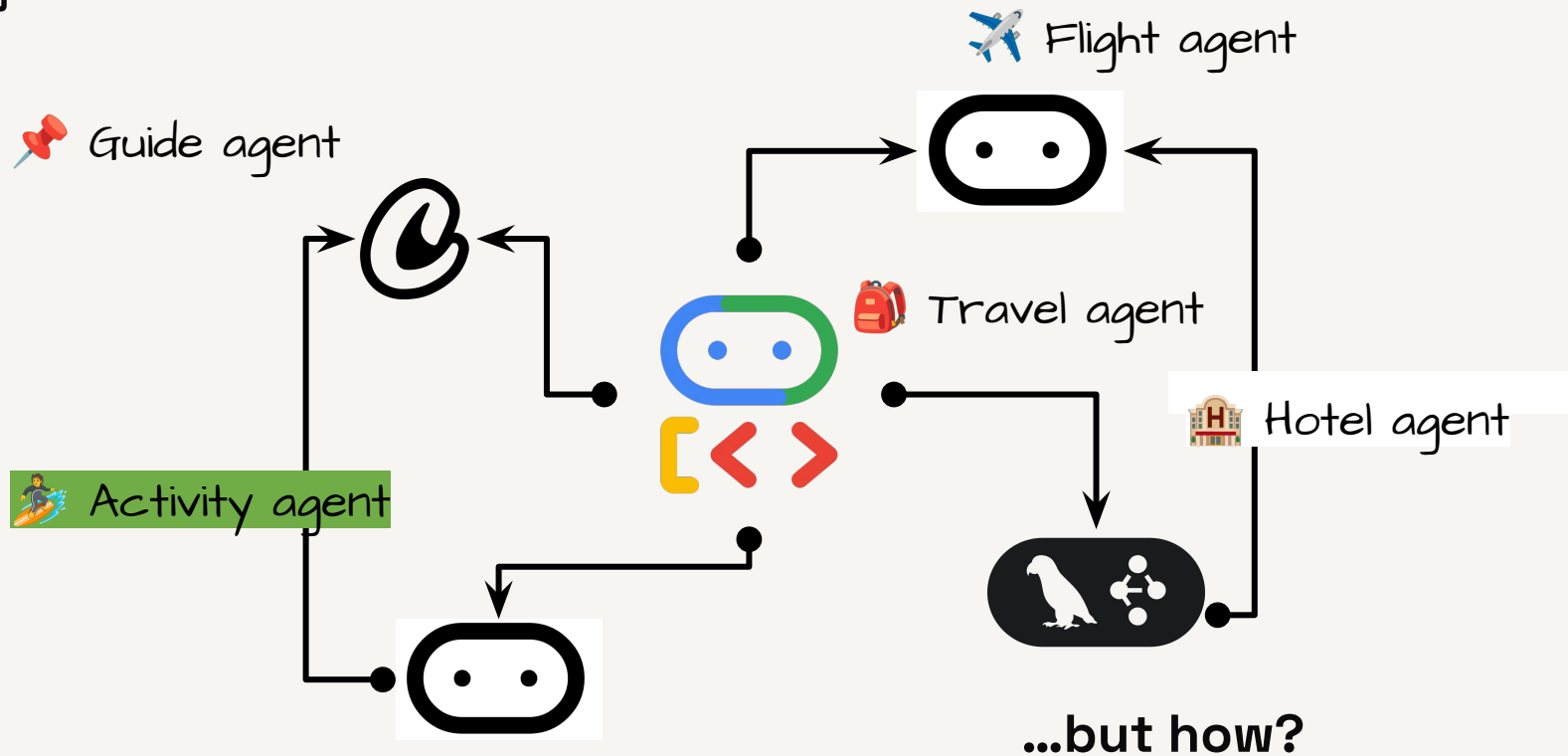
Sr. Cloud Architect · @Google

PART 01

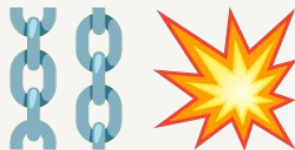
Introduction

What and why?

Agents need to talk!



Agents' challenges



Fragmented landscape

How do we achieve faster development and iteration?

Hard to integrate

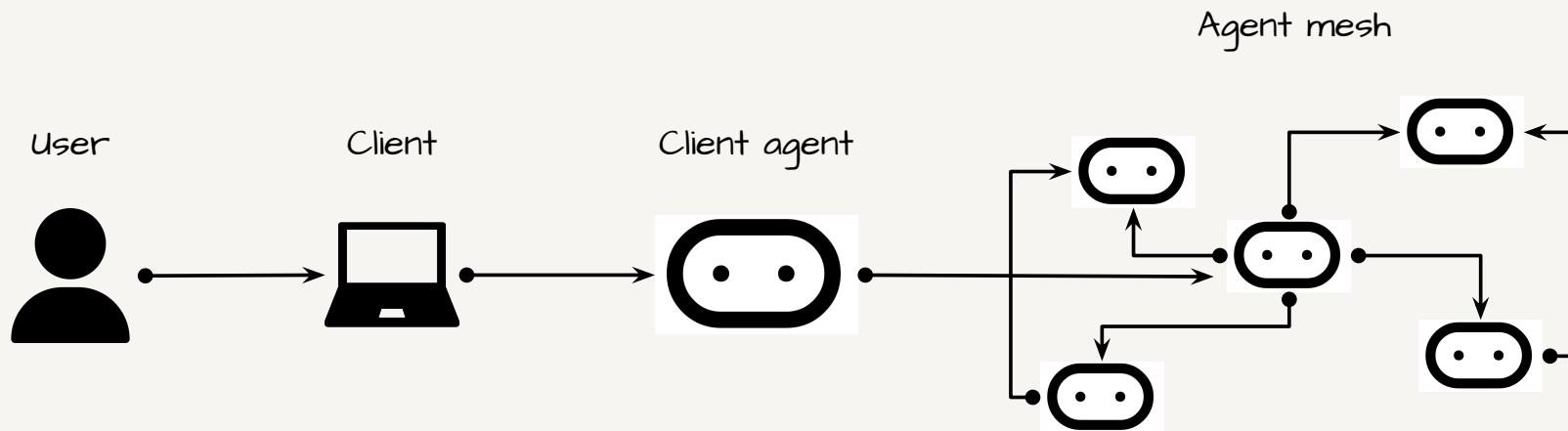
How do we manage dependencies, delegate effectively, and avoid creating hard-to-debug workflows between agents?

Lack of ops & governance

Beyond development, how do we reliably deploy, scale, monitor, and govern an ecosystem of agents?

Agent2Agent (A2A) protocol

Open protocol to handle agent collaboration



An open protocol started at Google, now part of the Agentic AI Foundation (AAIF), that standardizes how agents running on diverse frameworks and platforms communicate.

a2a-protocol.org

A2A capabilities



Discovery

Agents must advertise their capabilities so clients know when and how to utilize them for specific tasks.



Negotiation

Clients and agents need to agree on communication methods like text, forms, iframe, or audio/video to ensure proper user interaction.



Task and State Management

Clients and agents need mechanisms to communicate task status, changes, and dependencies throughout task execution.



Collaboration

Clients and agents must support dynamic interaction, enabling agents to request clarifications, information, or sub-actions from client, other agents, or users.

PART 02

Libraries and tools

SDKs, samples and the A2A Inspector

A2A on GitHub



github.com/a2aproject - GitHub Organization



github.com/a2aproject/A2A - Protocol and Documentation



github.com/a2aproject/a2a-python - Python SDK



github.com/a2aproject/a2a-js - TypeScript/Node.js SDK



github.com/a2aproject/a2a-java - Java SDK



github.com/a2aproject/a2a-go - Go SDK



github.com/a2aproject/a2a-dotnet - C#/.NET SDK



github.com/a2aproject/a2a-samples - Code Samples/Demos



github.com/a2aproject/a2a-inspector - Test A2A Agents

A2A SDK

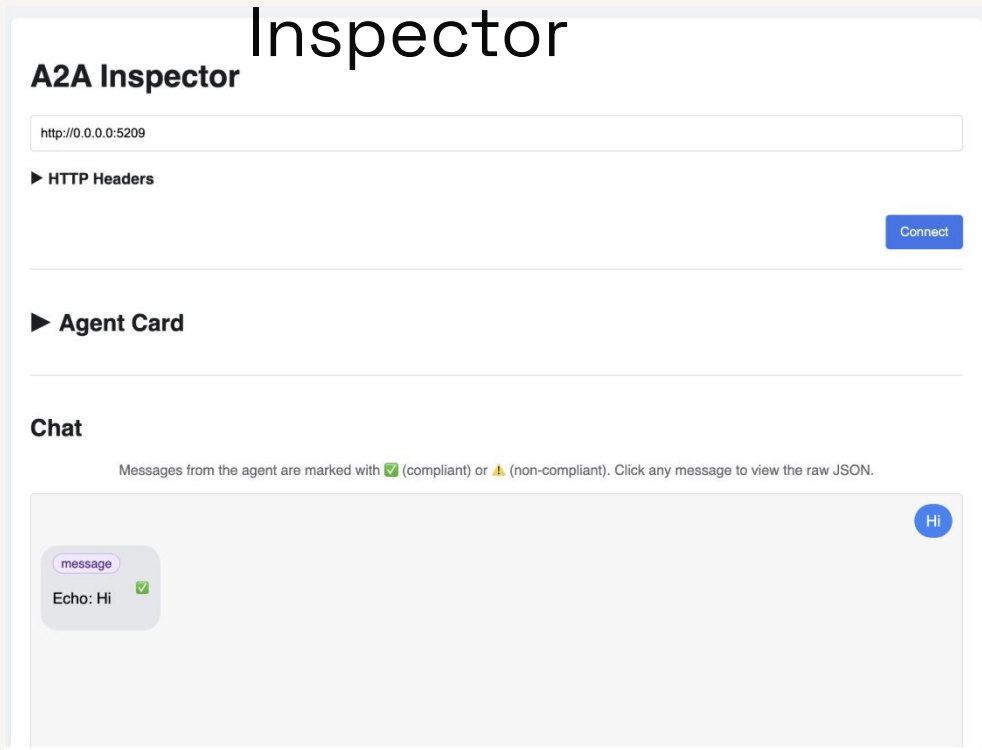
goo.gle/a2a-python-sdk

```
$ pip install a2a-sdk
```

```
from a2a.types import AgentCard, AgentCapabilities, AgentInterface, AgentSkill

agent_card = AgentCard(
    name='Hello World Agent',
    description='Just a hello world agent',
    version='1.0.0',
    supported_interfaces=[AgentInterface(
        url='http://localhost:9999/', protocol_binding='JSONRPC', protocol_version='1.0',
    )],
    capabilities=AgentCapabilities(streaming=True),
    default_input_modes=['text/plain'],
    default_output_modes=['text/plain'],
    skills=[skill],
)
```

A2A Inspector



github.com/a2aproject/a2a-inspector

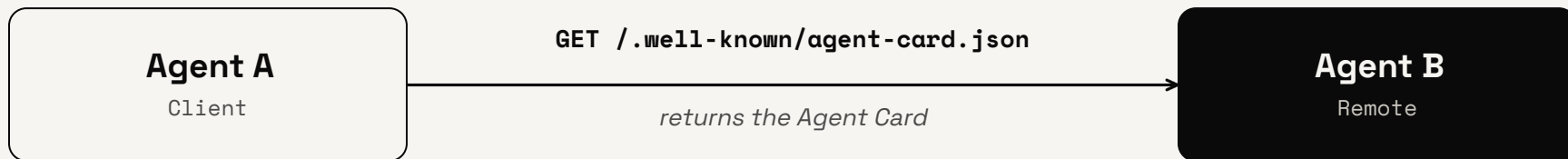
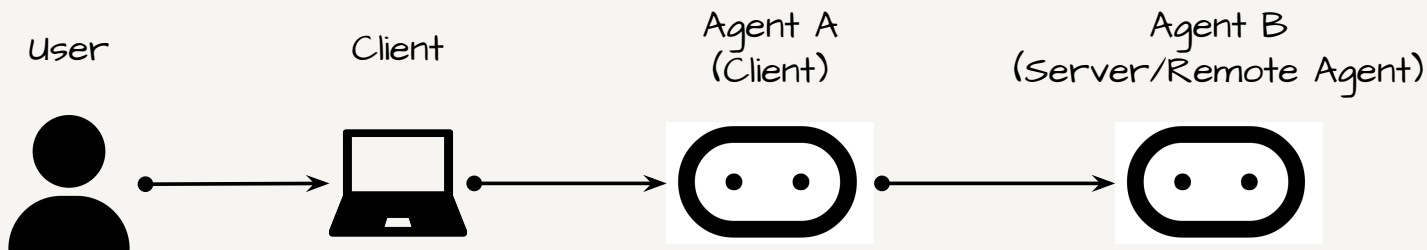
PART 03

How A2A works

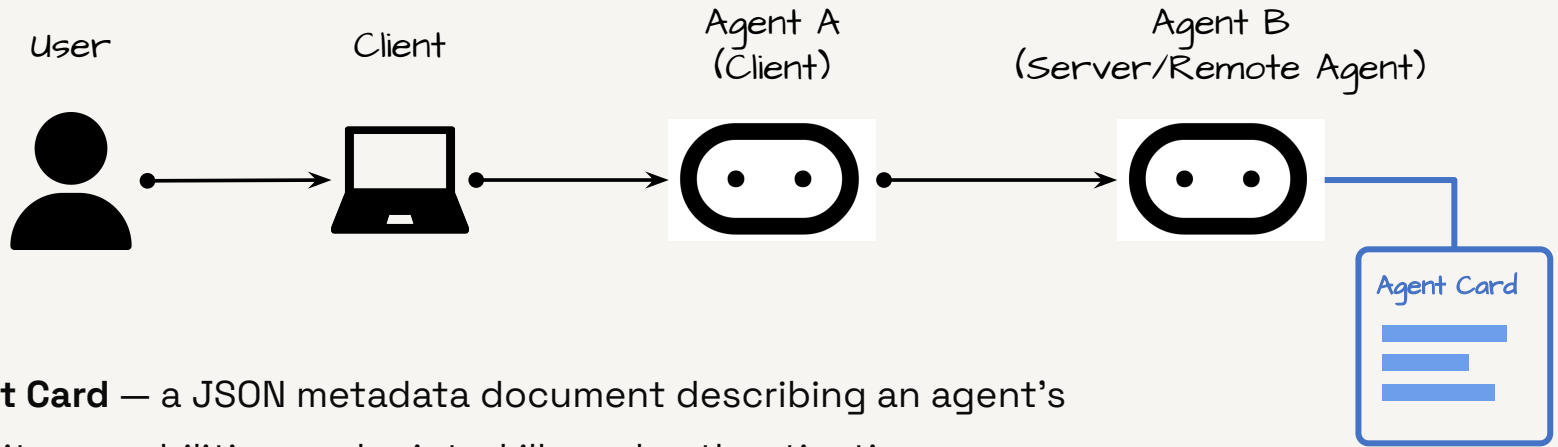
Discovery · AgentExecutor · Interactions

Agent Discovery

Who are you & what can you do?



The Agent Card



Agent Card — a JSON metadata document describing an agent's identity, capabilities, endpoint, skills and authentication requirements.

What an Agent Card looks

v1.0 JSON

```
{
  "name": "Currency Agent",
  "description": "Helps with exchange rates",
  "version": "1.0.0",
  "supportedInterfaces": [{
    "url": "http://127.0.0.1:9999",
    "protocolBinding": "JSONRPC",
    "protocolVersion": "1.0"
  }],
  "capabilities": { "streaming": true, "pushNotifications": true },
  "defaultInputModes": ["text/plain"],
  "defaultOutputModes": ["text/plain"],
  "skills": [{ "id": "get_exchange_rate", ... }],
  "securitySchemes": { ... }
}
```

Where to reach it

One entry per transport

What it supports

Streaming, push notifications

What it can do

Skills with examples

How to authenticate

OAuth, OIDC, API key, mTLS

AgentSkill

```
skill = AgentSkill(  
    id='get_exchange_rate',  
    name='Currency Exchange Rates Tool',  
    description='Gets exchange rates between currencies',  
    input_modes=['text/plain'],  
    output_modes=['text/plain'],  
    tags=['currency conversion', 'currency exchange'],  
    examples=['What is 1 GBP in USD?'],  
)
```

AgentCard - Example

a2a-sdk v1.0

```
agent_card = AgentCard(  
    name='Currency Agent',  
    description='Helps with exchange rates for currencies',  
    version='1.0.0',  
    supported_interfaces=[  
        AgentInterface(  
            url=f'http://{host}:{port}/',  
            protocol_binding='JSONRPC',  
            protocol_version='1.0',  
        ),  
    ],  
    capabilities=AgentCapabilities(  
        streaming=True, push_notifications=True,  
    ),  
    default_input_modes=['text/plain'],  
    default_output_modes=['text/plain'],  
    skills=[skill],  
)
```

NEW IN V1.0

url → supported_interfaces

One card, many transports: JSON-RPC, gRPC,
REST

Version per interface

protocol_version lives on each interface

snake_case in Python

default_input_modes, not defaultInputModes

PART 04

Agent interactions

How can Agent A interact with Agent B?

Agent response: message or task

Option 1

Respond with a stateless message

For trivial, immediate answers.

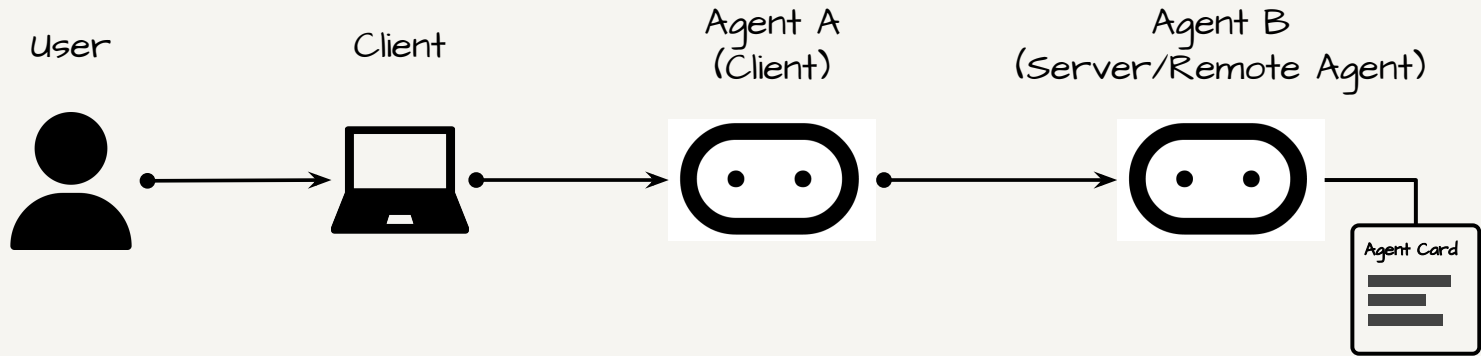
Option 2

Initiate a stateful task

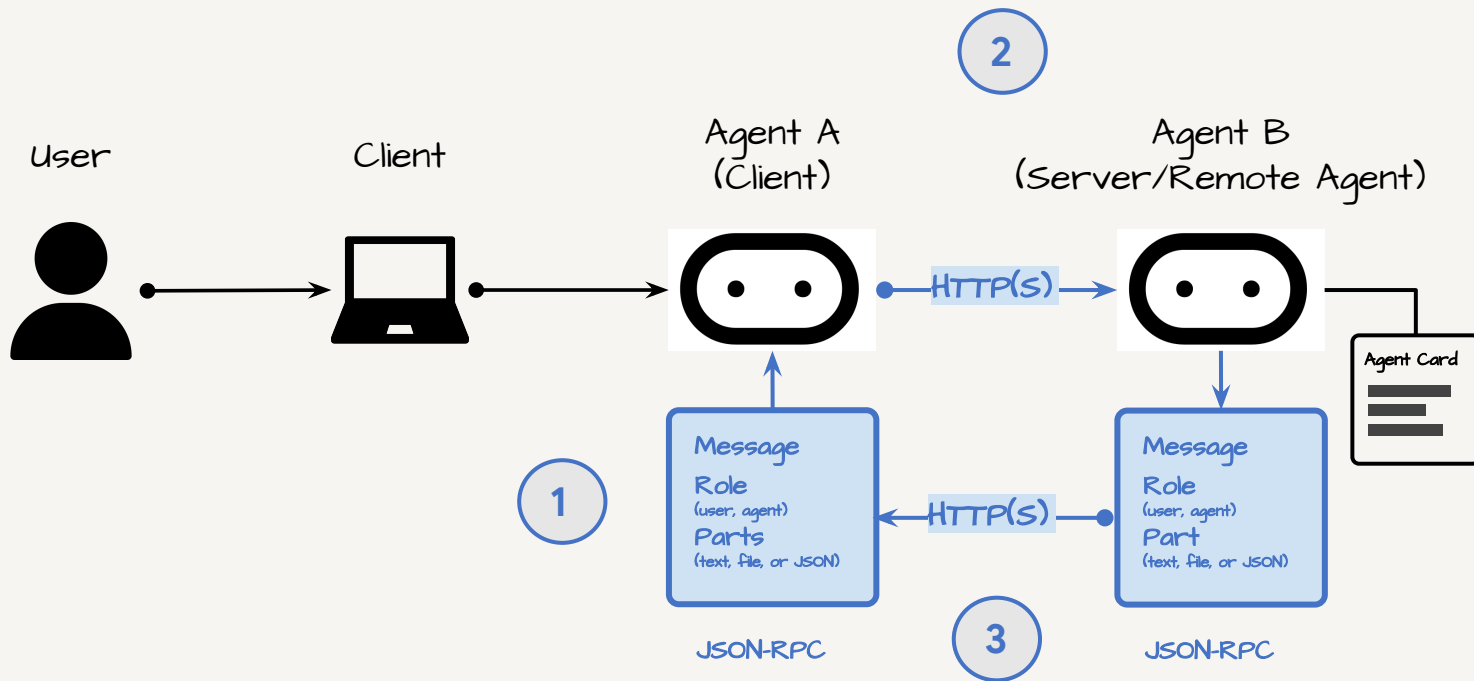
For real work that takes time and has a lifecycle.

Basic Interaction

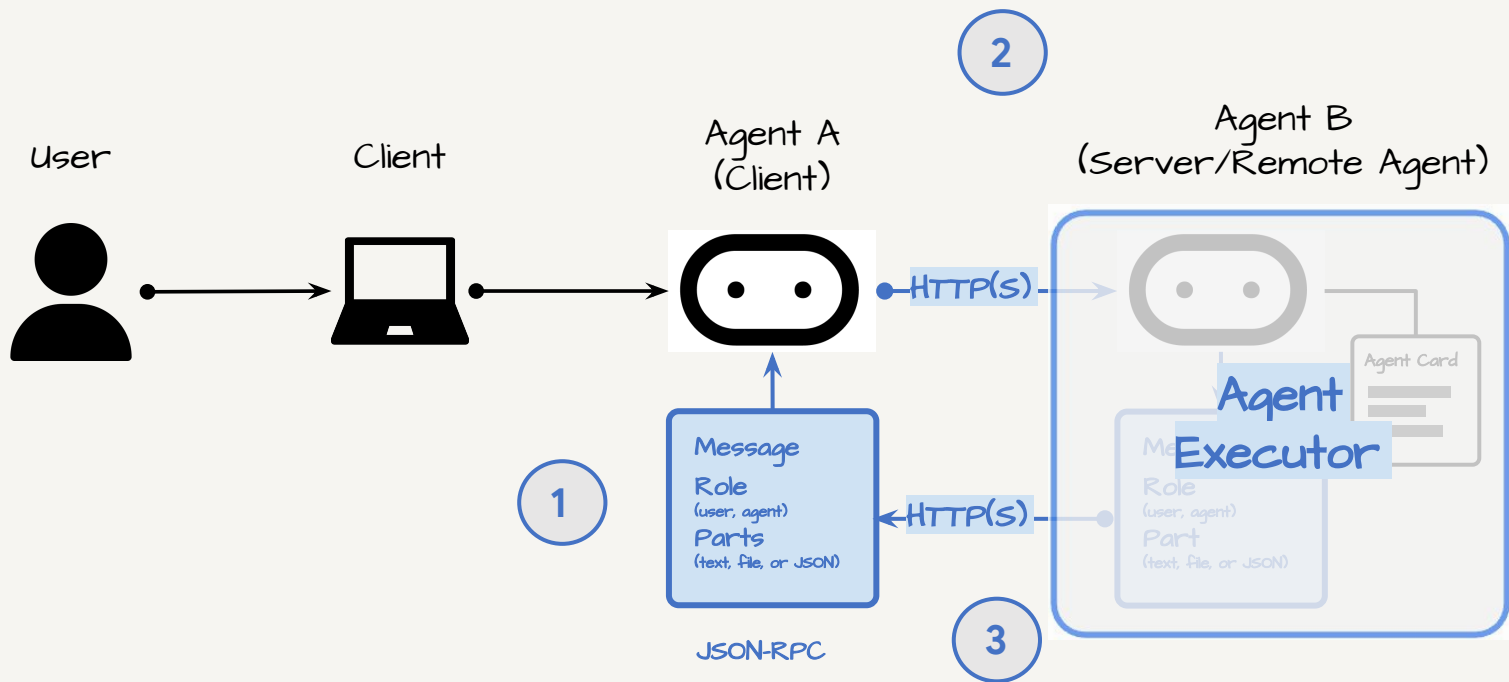
How do we actually talk?



Messages, Tasks & Parts



The Agent Executor

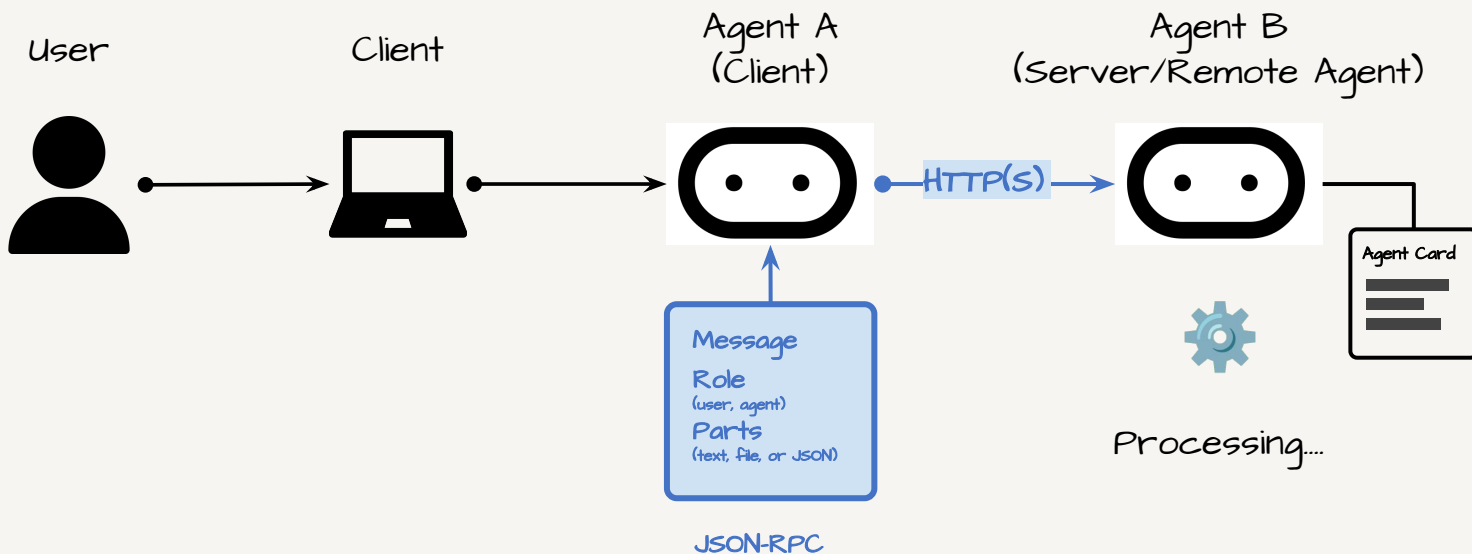


The AgentExecutor interface

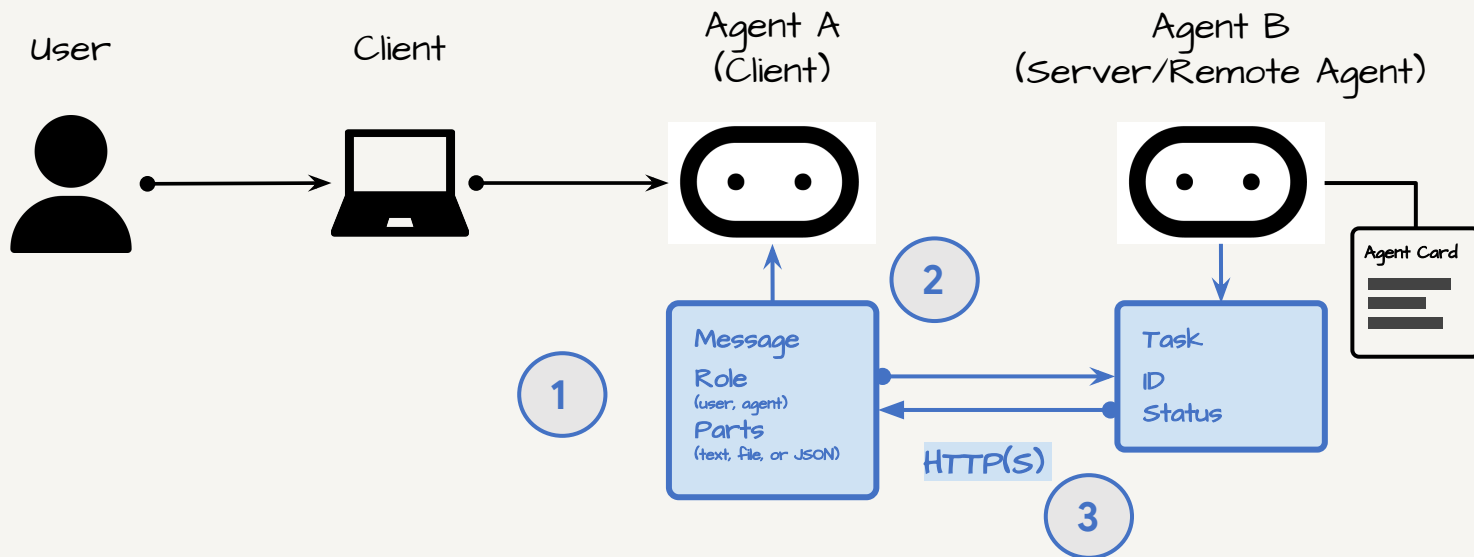
```
class AgentExecutor(ABC):  
    """Core logic of the agent: executes tasks and publishes updates to an event queue."""  
  
    @abstractmethod  
    async def execute(self, context: RequestContext, event_queue: EventQueue) -> None:  
        """Execute the agent's logic for a given request context."""  
  
    @abstractmethod  
    async def cancel(self, context: RequestContext, event_queue: EventQueue) -> None:  
        """Request the agent to cancel an ongoing task."""
```

Handling Real Work

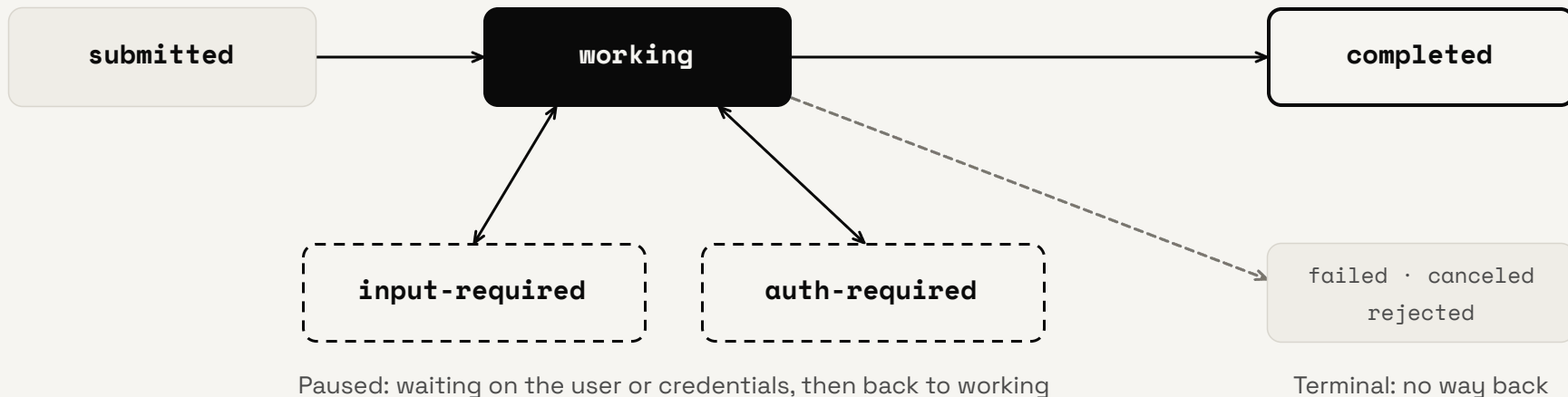
Are we there yet?



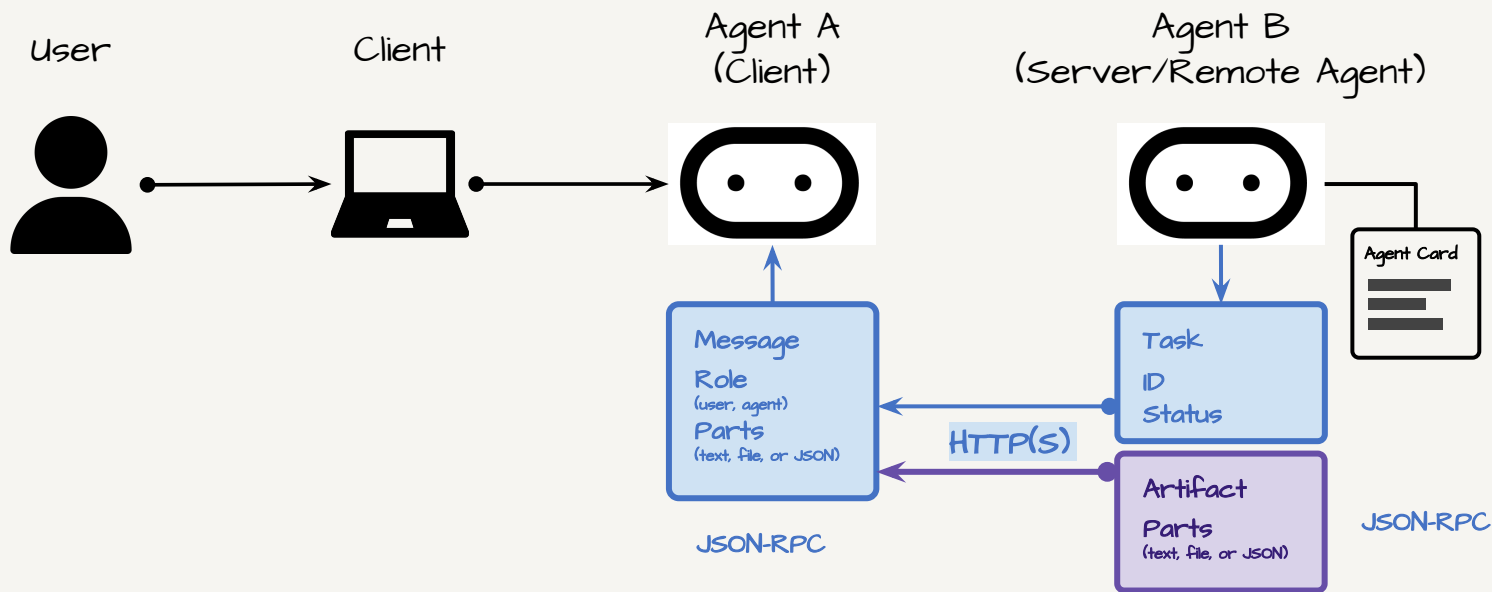
Task Lifecycle & Polling



Every task has a state

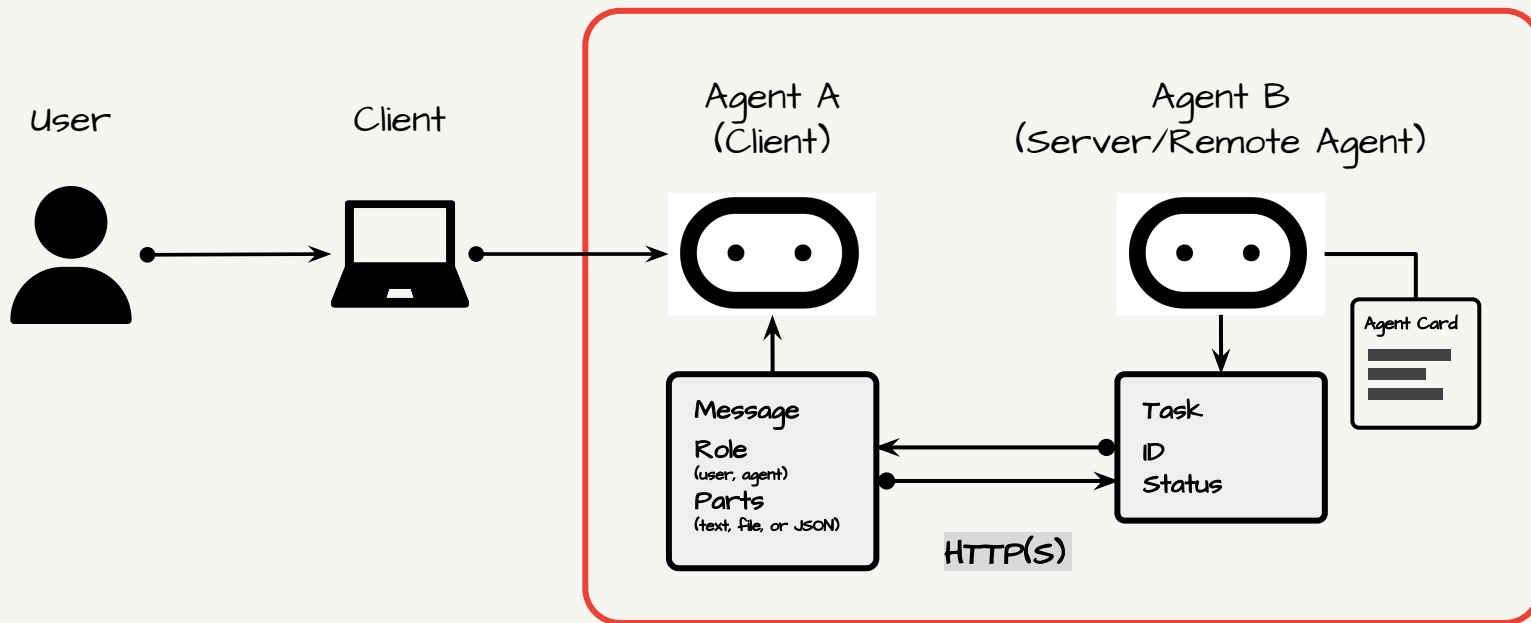


Task Lifecycle & Polling

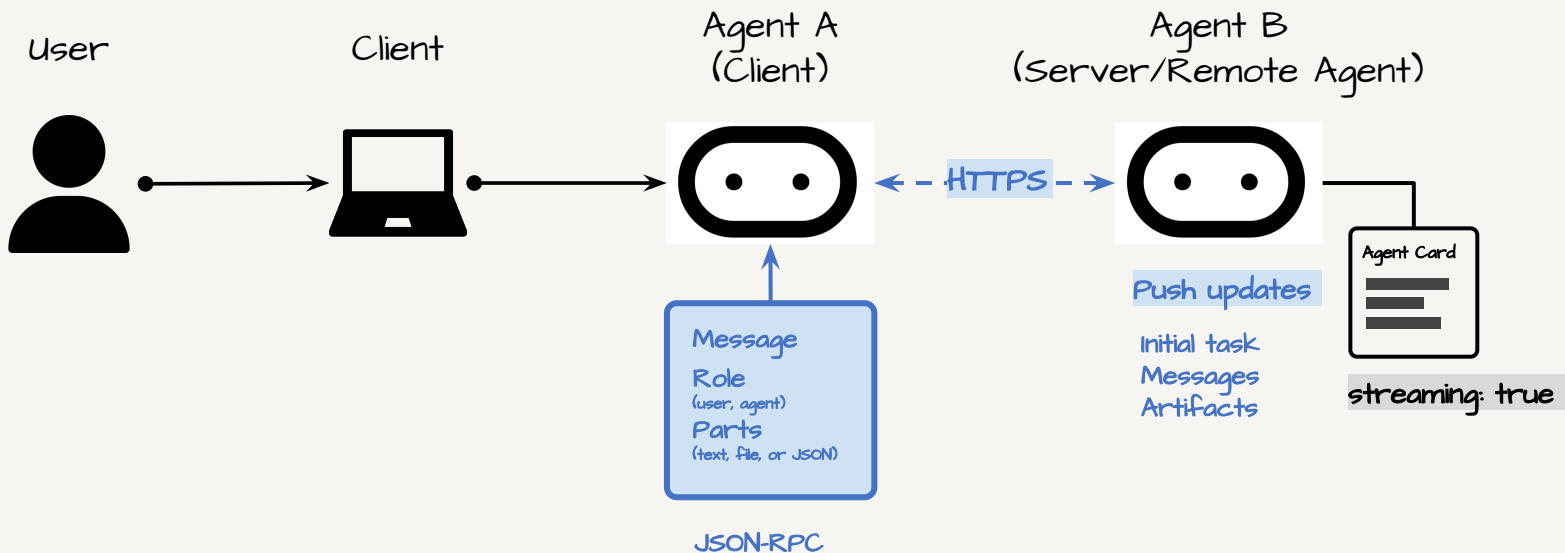


Task Lifecycle & Polling Challenge

✗ Polling is inefficient!



Real-time Updates Streaming with SSE



Streaming, multi-turn

a2a-sdk v1.0

```
async for item in self.agent.stream(query, task.context_id):
    if item['require_user_input']:
        await task_updater.update_status(
            state=TaskState.TASK_STATE_INPUT_REQUIRED,
            message=new_text_message(item['content']))
        return
    if item['is_task_complete']:
        await task_updater.add_artifact(
            parts=[new_text_part(text=item['content'])])
        await task_updater.update_status(
            state=TaskState.TASK_STATE_COMPLETED)
        return
    await task_updater.update_status(
        state=TaskState.TASK_STATE_WORKING,
        message=new_text_message(item['content']))
```

THREE BRANCHES

input-required

Pause and ask the user

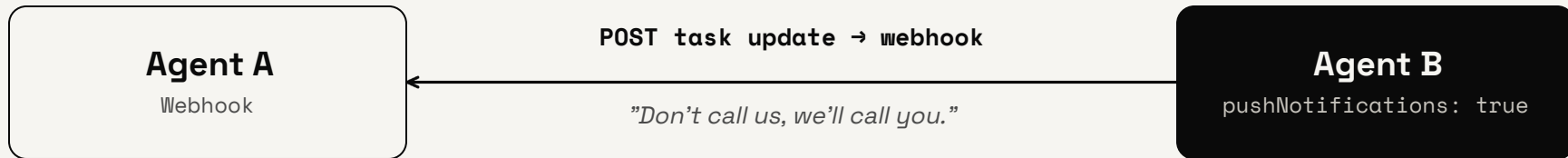
completed

Attach the artifact, finish

working

Stream progress updates

Push notifications for disconnected clients



BEST SUITED FOR

- Very long-running tasks: minutes, hours or days
- Clients that can't hold a persistent connection, such as mobile apps
- Clients that only need significant state changes, not every update

Send a non-streaming message

```
from a2a.client import ClientConfig, create_client
from a2a.helpers import new_text_message
from a2a.types import Role, SendMessageRequest

config = ClientConfig(streaming=False)
client = await create_client(agent=public_agent_card, client_config=config)

message = new_text_message('Why is the sky blue?', role=Role.ROLE_USER)
request = SendMessageRequest(message=message)
async for chunk in client.send_message(request):
    print(chunk)
```

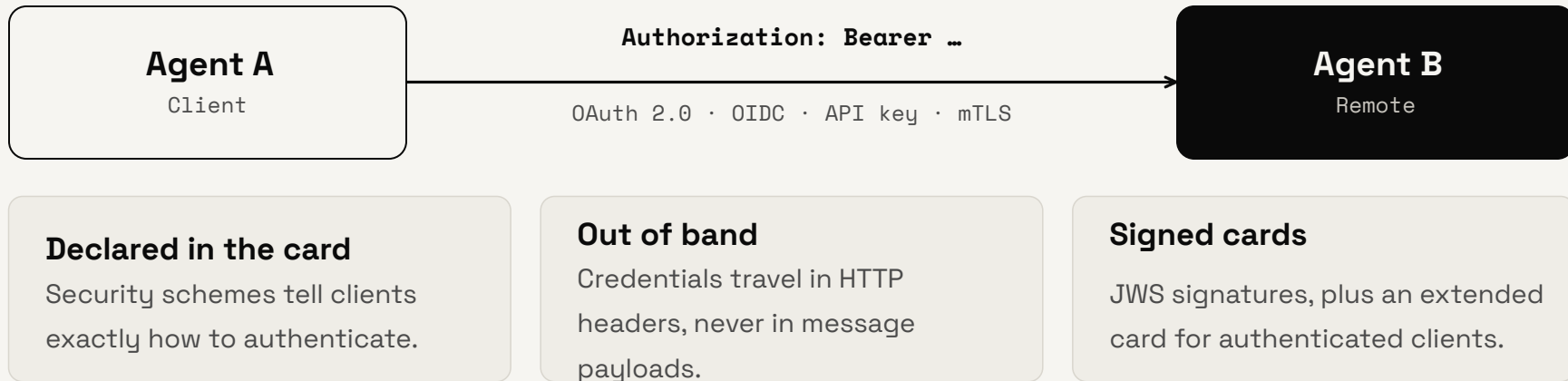
PART 05

Production reality

Who's calling, who's allowed, and what happened?

SECURITY

Trust: who's calling?



GOVERNANCE

Govern the A2A boundary



Who is this agent?

Identity

A unique identity per agent (e.g. SPIFFE), not a shared service account.



Who may call whom?

Policy

Gateway allow-lists and egress rules at every A2A hop.



What happened?

Observability

OpenTelemetry traces that follow one task across every agent.

A2A gives you interoperability. **Governance gives you trust.**

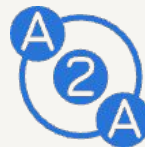
A2A and MCP?

Complementary, Not Competing



Model Context Protocol (MCP)

- Connects agents to **tools, APIs, and resources**.
- Think: *How an agent uses its capabilities (function calling).*
- Example: Agent uses MCP to call a weather API tool.



Agent2Agent Protocol (A2A)

- Facilitates dynamic communication **between different agents** as peers.
- Think: *How agents collaborate, delegate, and manage shared tasks.*
- Example: A Travel Agent (A2A) asks a Flight Booking Agent (A2A) to find flights.



GET STARTED

Where to learn more



Read the docs

<https://a2a-protocol.org>



Get the code

[github.com/a2aproject/
a2a-samples](https://github.com/a2aproject/a2a-samples)



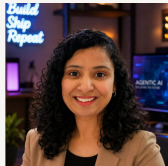
Try ADK + A2A

goo.gle/adk-a2a

**Demo - [Multi-Agent Systems
with Agent2Agent](#)**

04

“Agent Observability”

**Jothsna Praveena**

Data Scientis · @Infosys @DeepAgentLabs

**Pramod B N**

Independent Researcher · @DeepAgentLabs

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Thank you

TONIGHT'S HOST

UTD. — *Facility, A/V***UTD AI Society** - *Facilitation, and MC***ZScaler** - *Food & Beverages*

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